THE IMAGE OF THE COLIMIT OF THE LOCAL COHOMOLOGIES ALONG THE PRINCIPAL OPENS THROUGH A PRIME IS THE SET OF ELEMENTS DYING AT THAT PRIME

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-local-cohomology/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The local cohomology spectral sequence for the two supports in question, in the form recorded in the chapter on dualizing complexes. Invoked inside the counted proof at source lines 473-475 as a cross-chapter pointer. Degenerates to the displayed short exact sequences that identify the pieces of the colimit.
- The vanishing of the colimit of the local cohomology of the principal opens of a multiplicative set for a module localised to zero at the prime. Author lemma lemma-multiplicative, whose statement is reproduced verbatim as a context span of this card (source lines 434-440). Supplies the second half of the description of the image in assertion (2).

Source reference. Original reference dualizing-lemma-local-cohomology-ss: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 428–428. Navigation pointer only: the counted text refers to the Stacks item dualizing-lemma-local-cohomology-ss, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $I \subset A$ be a finitely generated ideal of a ring A. Let $\mathfrak{p}$ be a prime ideal. Let M be an A-module. Let $i \geq 0$ be an integer and consider the map*

$$\Psi : \operatorname{colim}_{f \in A, f \notin \mathfrak{p}} H^i_{V((I,f))}(M) \longrightarrow H^i_{V(I)}(M)$$

Then

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(1) $\mathrm{Im}(\Psi)$ *is the set of elements which map to zero in* $H^i_{V(I)}(M)_\mathfrak{p}$,
(2) *if* $H^{i-1}_{V(I)}(M)_\mathfrak{p} = 0$, *then* Ψ *is injective,*
(3) *if* $H^{i-1}_{V(I)}(M)_\mathfrak{p} = H^i_{V(I)}(M)_\mathfrak{p} = 0$, *then* Ψ *is an isomorphism.*

Lemma 0.2. *Let* $S \subset A$ *be a multiplicative set of a ring* A. *Let* M *be an* A*-module with* $S^{-1}M = 0$. *Then* $\mathrm{colim}_{f\in S} H^0_{V(f)}(M) = M$ *and* $\mathrm{colim}_{f\in S} H^1_{V(f)}(M) = 0$.

Proof. For $f \in A$, $f \notin \mathfrak{p}$ the spectral sequence of Dualizing Complexes, Lemma dualizing-lemma-local-cohomology-ss degenerates to give short exact sequences

$$0 \to H^1_{V(f)}(H^{i-1}_{V(I)}(M)) \to H^i_{V((I,f))}(M) \to H^0_{V(f)}(H^i_{V(I)}(M)) \to 0$$

This proves (1) and part (2) follows from this and Lemma 0.2. Part (3) is a formal consequence. □